

Md. Modacher Mahmud Rafi

📍 Dhaka Division, Bangladesh ✉ work.rafi.mahmud@gmail.com ☎ +880 1799-951107 🌐 in/modacher-mahmud-rafi

☁ modachermahmudrafi-one.vercel.app

Summary

Dynamic Junior Flutter Developer with 1.5+ year of experience at Sparktech Agency, specializing in Flutter, Firebase, and API integration. Proven ability to build high-performance, pixel-perfect apps with improved load times and user engagement. Software Engineering graduate passionate about real-time mobile applications and continuous learning.

Experience

Sparktech Agency

Jr. Flutter Developer

Mohakhali, Dhaka

February 2025 - Present

- **Flutter/Dart** development of parcel delivery, Time Tracker (audio recording/processing), and Salon booking apps across **iOS and Android** platforms; integrated Firebase (Auth, Firestore, Cloud Functions) for **real-time data** synchronization and authentication, reducing load times 30% via optimized queries.
- Delivered responsive, pixel-perfect UIs from Figma/Adobe XD prototypes across **Android/iOS** devices; applied GetX, RxDart, Provider for scalable, reactive state management using **Dart** best practices.
- Integrated **RESTful APIs** and WebSockets to enable seamless, **real-time data** sync between mobile clients and backend services, elevating user engagement for high-traffic applications with instant updates and notifications.
- Implemented native platform channels using Kotlin (**Android**) and Swift (**iOS**) for custom functionality, ensuring optimal performance and native feature integration.

ICT Division of Bangladesh

Python Trainer

Dhaka, Bangladesh

January 2024 - January 2025

- Database Integration with **MySQL**: Guided trainees in connecting Python with **MySQL** databases, executing **SQL queries**, and managing data.
- Building Project : Helped participants develop basic console-based applications by integrating Python with database operations and also using **Tkinter** for GUI application.
- Fostered problem solving skills among 70+ trainees by designing weekly scenarios and hands-on projects using Python, MySQL, and Tkinter over a 6-month program, leading to measurable gains in successful task completion rates (25% increase tracked via learning analytics tools).

Skills

- **Framework:** Flutter
- **Language:** C++, Java, Python, Dart
- **State Management:** Get, RxDart, Provider
- **Backend:** Firebase (Firestore, Firebase Auth), Appwrite (Auth, Database, Function)
- **API Integration:** RESTful APIs, WebSockets, Real-time data synchronization
- **Version Management:** Git, Github
- **Architecture:** MVC, MVVM
- **Fields of Interest:** Learning Go, Chess, Machine Learning, Algorithms, Advanced Data Structures, Artificial Intelligence

Project

Sparktech Agency | Time Tracker Application

apps.apple.com/ca/app/alpha-track/id6757949847

July 2025 - September 2025

- Built Flutter Time Tracker app using **Dart** for employee project assignment and time monitoring across **iOS and Android** devices; owners set schedules/timers while employees view tasks, log hours, and access details via **RESTful API** backend integration.

- Implemented **real-time data** synchronization using WebSockets and **RESTful APIs** to enable instant updates for project assignments, timer status, and collaborative features.
- Added collaborative note section for owners/employees to attach documents, images, and audio files to projects, with extensive audio recording/processing features leveraging native **iOS/Android** capabilities.

Project Finder | *Project Finder*

May 2025 - October 2025

apps.apple.com/us/app/project-finder-app/id6754875552

- Engineered a scalable subscription infrastructure that seamlessly incorporated auto-renewal, comprehensive billing processes, and secure account-based management for an enhanced user experience.
- Devised an project discovery platform that matched users with relevant opportunities using advanced filtering and recommendation algorithms, elevating engagement and platform adoption rates.

Sparktech Agency | *Senaeya*

August 2025 - December 2025

play.google.com/store/apps/details?id=com.fahadalfayez.senaeya&hl=en-US

- Streamlined workshop operations by implementing intuitive workflows and delivering consistent enhancements that elevated usability and service quality.
- Engineered a scalable, all-in-one workshop management platform (Senaeya) supporting vehicle registration, invoicing, reporting, and expense tracking, which improved operational efficiency by 35% across 500+ workshops while enabling seamless mobile access and eliminating the need for additional hardware.

Sparktech Agency | *Salon Go*

May 2025 - August 2025

play.google.com/store/apps/details?id=com.salon.go&hl=en-US

- Built cross-platform Flutter app in **Dart** for **iOS and Android** with mobile sign-in, salon booking tracking, and Google Maps integration for routes, location suggestions, and interactive maps.
- Integrated **RESTful APIs** for booking management, user authentication, and service catalog with seamless **real-time data** updates across platforms.
- Added Firebase push notifications and WebSockets for **real-time** parcel location/status updates, enabling instant tracking and synchronized status alerts during send/delivery with continuous **real-time data** streaming.

Education

Bachelor of Software Engineering Major in Data Science

Daffodil Smart City, Birulia, Ashulia, Savar
2024

Daffodil International University • 3.65

- Escape the Room Contest - 2021 - Position : 31st
- Unlock the Algorithm - 2021 - Position : 38th
- Unlock the Algorithm - 2022 - Position : 47th
- Participate at 3rd & 4th DIU Data Science Summit - 2022, 2024

Involvement

Daffodil International University

- **DIU CPC Workshops:** Attended DIU CPC workshops and training, enhancing skills related to computer programming and competitive coding
- **Chess Club Member:** Actively participated as a member of the Daffodil International University Chess Club, engaging in strategic gameplay and club activities.

Publications

A Machine Learning Approach to Predict Flood in Bangladesh from Historical Data | *Thesis*

Daffodil International University Journal (D Space) • 2024

- Data collected from the Meteorological Department of Bangladesh.
- Using Regression Approach to Predict the flood different parameter
- Using their ensemble learning different regression types of models.